



BESS Power Availability Deep Dive

How to measure and compute your BESS ability to
Deliver/Discharge or Absorb/Charge Power

Module 1 - BESS Availability Explained

 by Fred Rivollier



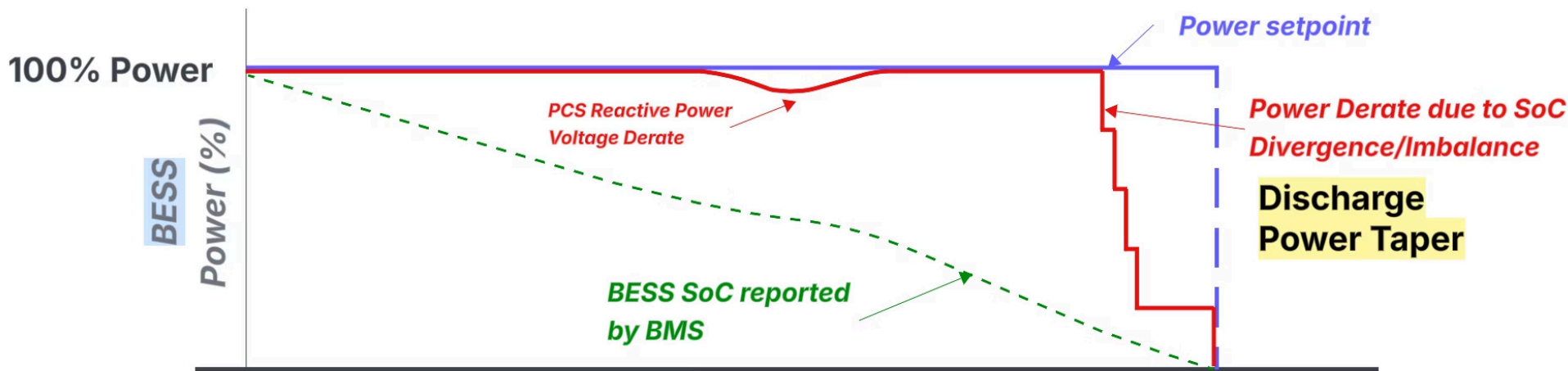
BESS Availability Types Recap

Combining all 4 availability type will form **OBE (Overall Battery Effectiveness)**

<u>Availability Type</u>	<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
① Equipment Availability	MEDIUM (ON/OFF)	HIGH 100% of Calendar Time	HIGH (Equipment Status signals)	HIGH Imbalance, SoC error, Derates	%	MW_installed MW_contracted
② Operational Availability	HIGH (MW)	LOW Only During Dispatch	HIGH (Meter Measurement/ Setpoint)	HIGH All time with dispatch command < nameplate power	MW	%_contracted
③ Power Availability	HIGH (MW)	HIGH 100% of Calendar Time	MEDIUM Calculated by EMS. Based on various equipment reported metrics	MEDIUM Many corner case and special condition that might be hard to model.	MW	%_installed %_contracted
④ Energy Availability	HIGH (MWh)	MEDIUM 100% of Calendar Time but might only be updated with capacity tests annually	LOW to MEDIUM Calculated by EMS, include SoC Error and Imbalance	MEDIUM Might include power derated energy due to imbalance	MWh	%_installed %_contracted

③ Power Availability

<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
HIGH (MW)	HIGH 100% of Calendar Time	MEDIUM Calculated by EMS. Based on various equipment reported metrics	MEDIUM Many corner case and special condition that might be hard to model such as reactive power and temperature derating.	MW	%_installed %_contracted



Available Power Derate Potential Causes



Equipment Unavailability

Equipment ON/OFF status directly reduces available power capacity

(Overlapping with Equipment Availability but not a replacement for Equipment Availability - different nominal capacity)



SoC Imbalance

Some strings are empty or full before others, limiting overall DC Bus power output



BMS Current Limiting

Temperature, balancing, and other BMS limits restrict current delivery



PCS Temperature

Ambient temperature power derate curves reduce PCS output



PCS AC Voltage

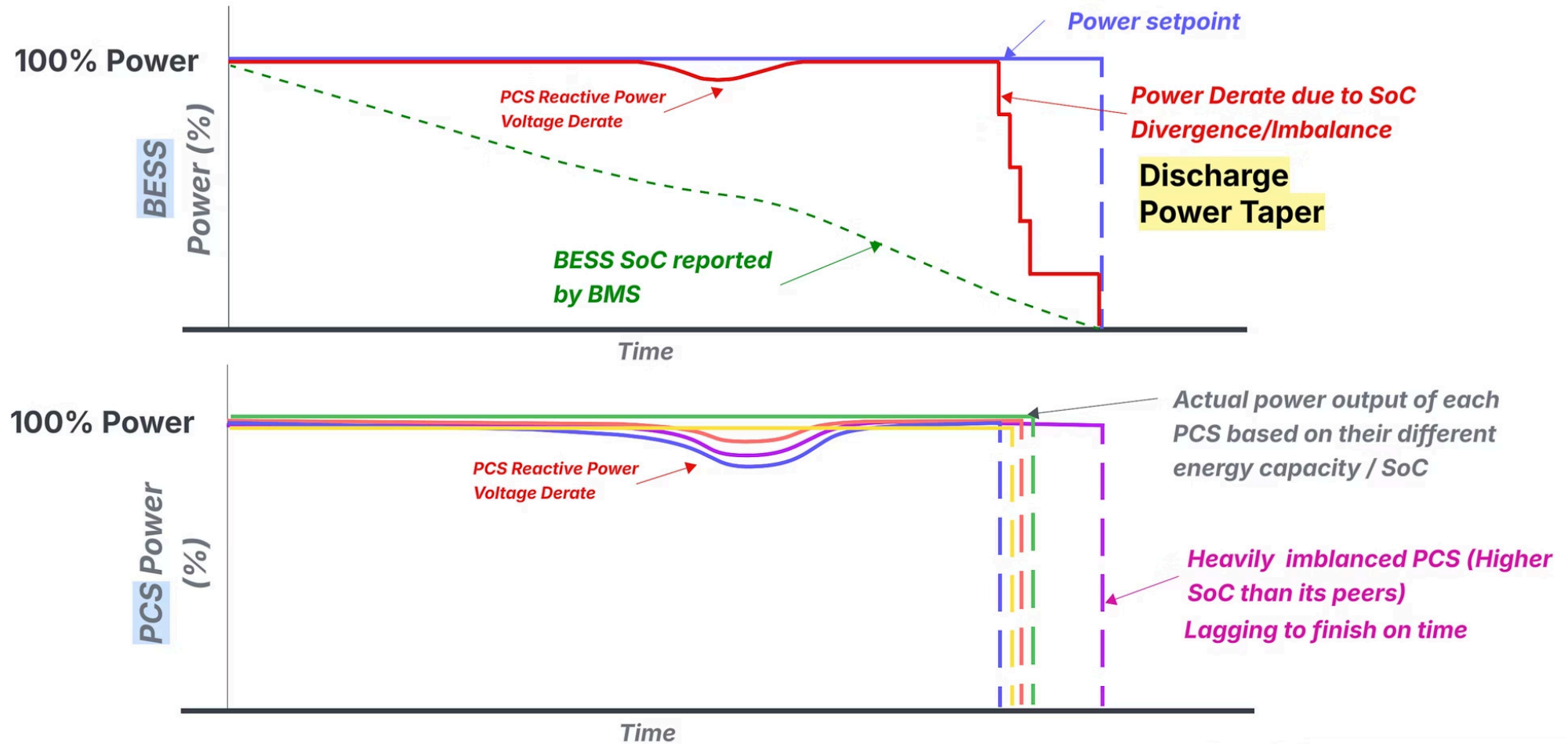
DC:AC conversion requires a certain voltage ratio to operate at full power



PCS Reactive Power

Apparent Power limit, reactive power priorities may limit Available Active Power

BESS Power Derating Examples



Normalization of Maximum Power Capacities

POI Maximum PCS Module in MW at POI:

PCS to POI losses have to be measured or estimated
est. to 2% loss for this example

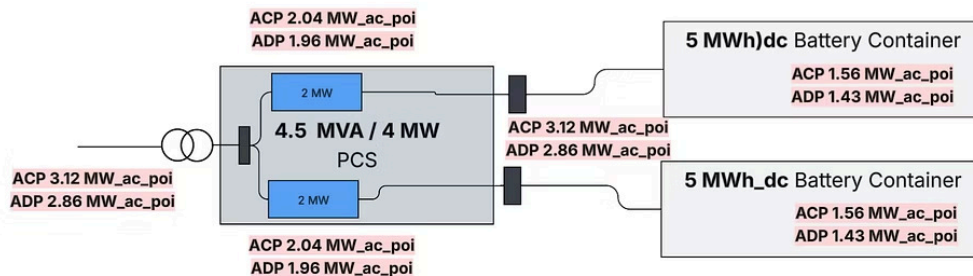
Make sure to account for Reactive Power requirement and AC line voltage range and always start from PCS MVA rating.

PCS 4.5 MVA = 4 MW active power usable.
(Account for Reactive Power + Line Voltage headroom)

Losses are helping available charge power

$$\text{PCS Module Maximum ACP} = \frac{1}{2} * (4 \text{ MW} * (1 + \text{POI Loss})) = 2.04 \text{ MW}$$

$$\text{PCS Module Maximum ADP} = \frac{1}{2} * (4 \text{ MW} * (1 - \text{POI Loss})) = 1.96 \text{ MW}$$



POI Maximum String Capacities in MW at POI:

Nominal C rate = 0.25

Maximum C rate = 0.3 (Current limit above nominal rated C rate)

4 h (0.25C rate) can operate at higher C rate to allow for balancing and partial availability compensation

Total DC-POI Loss = PCS Conversion Loss + POI loss

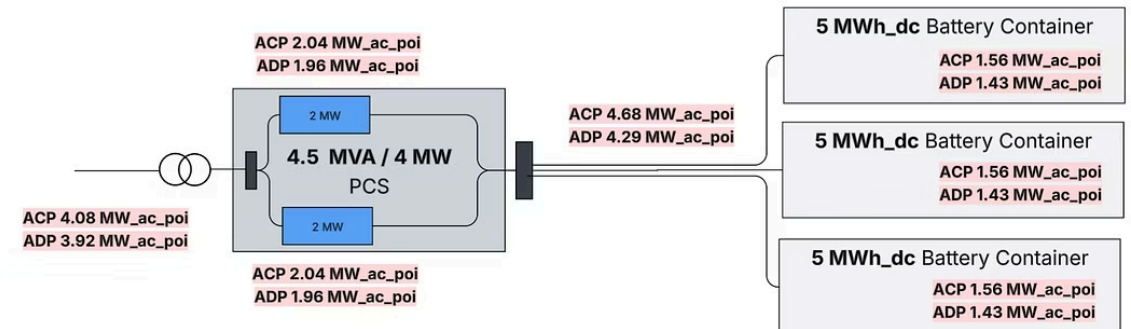
DC-POI Charge Losses +4.0%

DC-POI Discharge losses -4.5%

System can operate at higher C rate than nominal 4h equivalent.

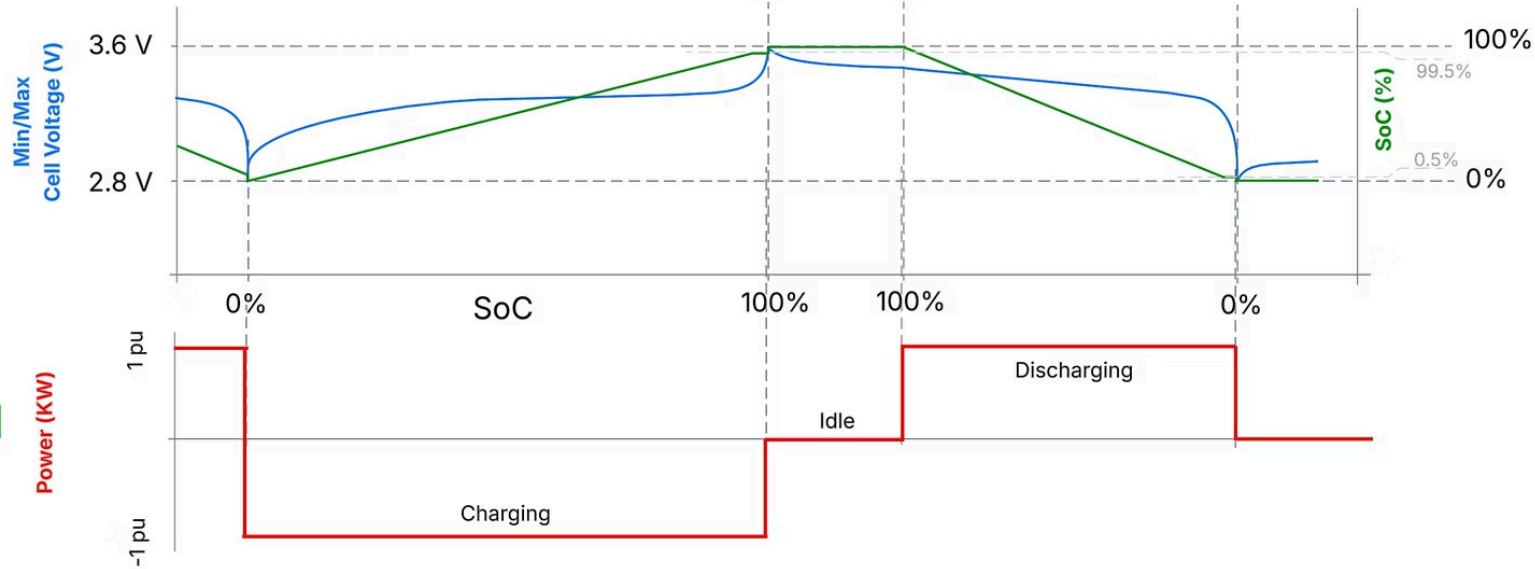
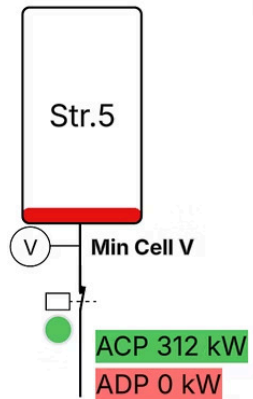
$$\text{String Maximum ACP} = 1 \text{ MWh} * (1 + \text{DC-POI Charge Loss}) * 0.3 = 0.312 \text{ MW}$$

$$\text{String Maximum ADP} = 1 \text{ MWh} * (1 - \text{DC-POI Discharge Loss}) * 0.3 = 0.286 \text{ MW}$$

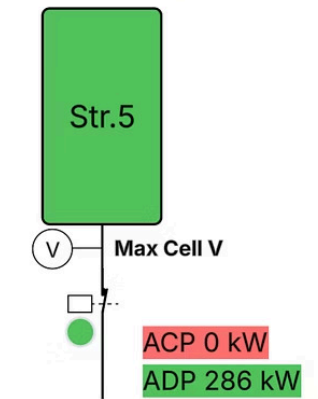


Battery String - Available Power based on SoC

0% SoC



100% SoC

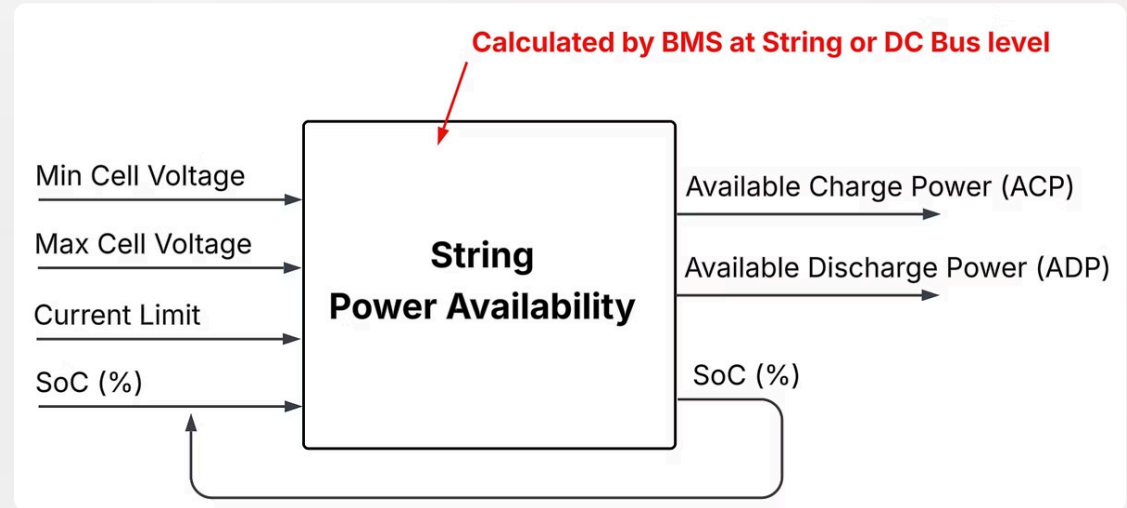


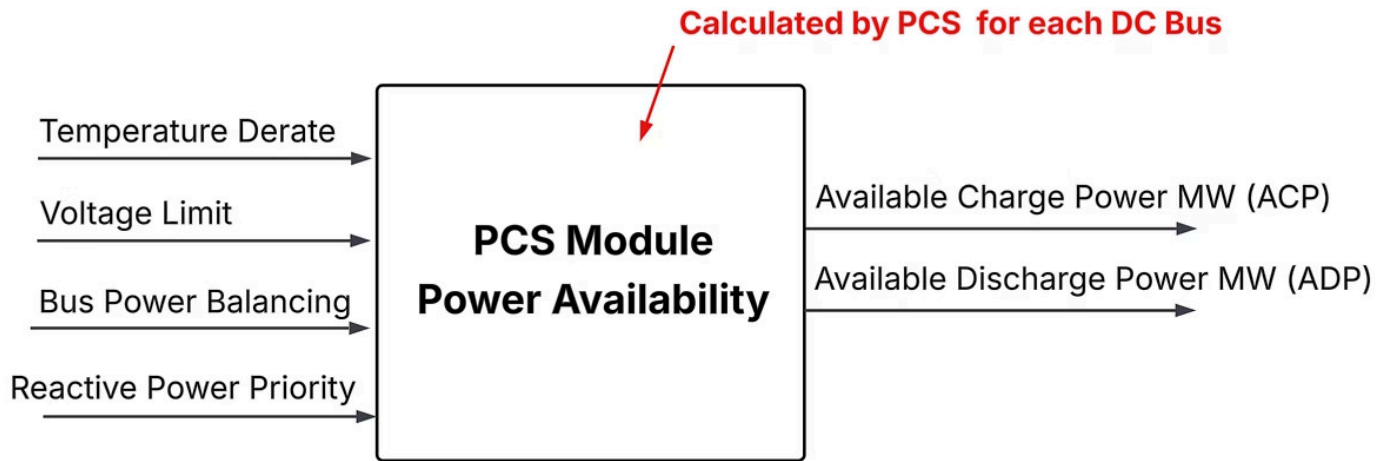
if SoC = 100% then: Available Charge Power (ACP) = 0 kW else: 312 kW_{ac_poi}

if SoC = 0% then: Available Discharge Power (ADP) = 0 kW else: 286 kW_{ac_poi}

String - Power Availability Function Block

String Available Power is best calculated by BMS at the DC Bus level by a Master BMS rather than String BMS.





PCS Module - Power Availability Function Block

PCS Module Available power should include charge and discharge efficiency to POI and be calculated by the PCS.

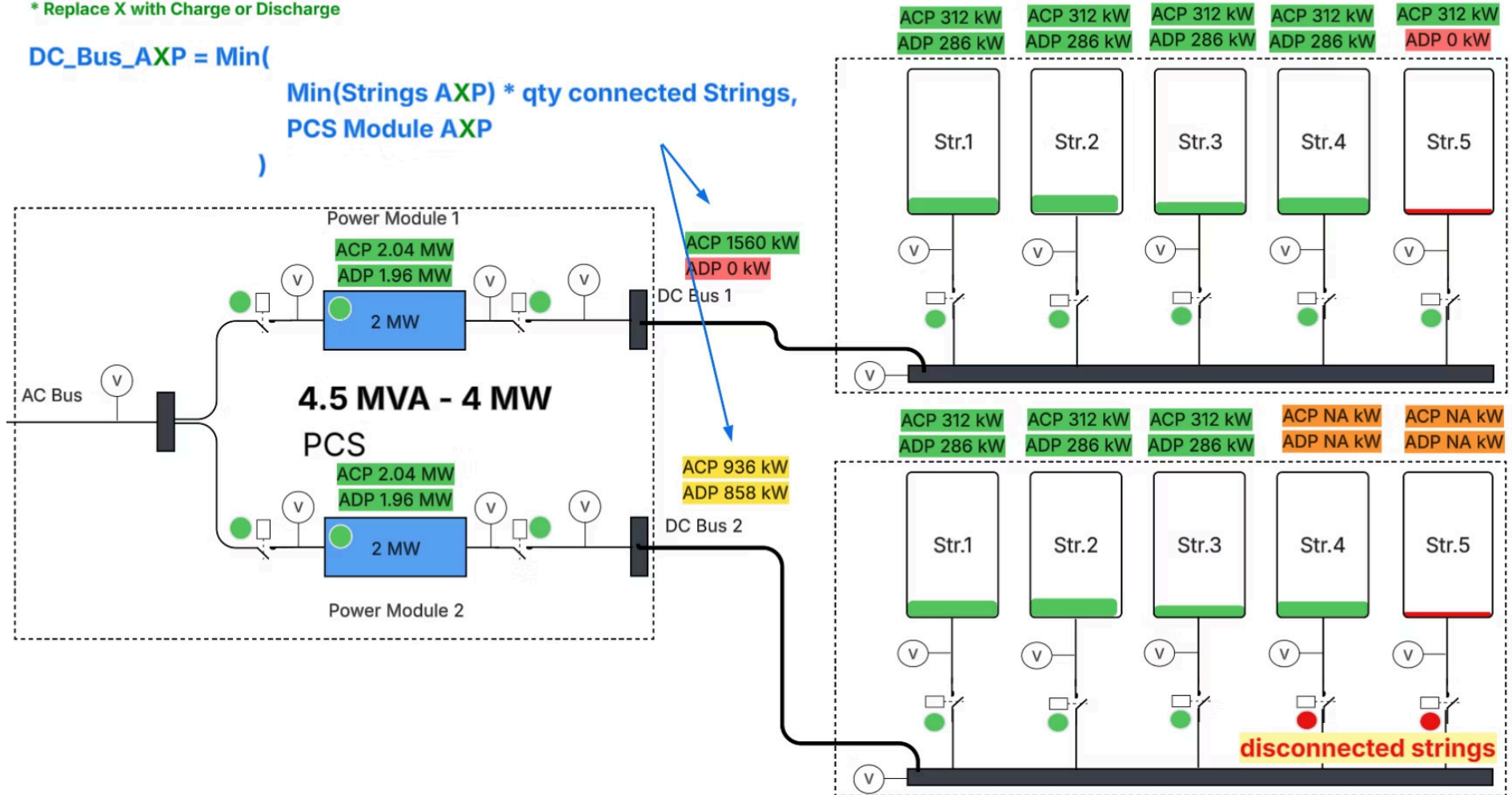
Available Power - Rolled up at DC Bus Level

* Replace X with Charge or Discharge

DC_Bus_AXP = Min(

Min(Strings AXP) * qty connected Strings,
PCS Module AXP

)



Availability — Charge, Discharge, Full Range & L



Available Power Calculation

* Replace X with Charge or Discharge



Bus Available Charge or Discharge Power (ACP, ADP)

$$\text{Bus_AXP} = \text{Min}(\text{Min}(\text{Strings_AXP}) \times \text{qty connected Strings}, \text{PCS_Module_AXP})$$



PCS Available Charge or Discharge Power

$$\text{PCS_AXP} = \text{Sum}(\text{Bus_AXP})$$



BESS Available Charge or Discharge Power

$$\text{BESS_AXP} = \text{Min}(\text{Sum}(\text{PCS_AXP}), \text{Contracted Power})$$



BESS Combined Available Power (AP)

$$\text{BESS_AP} = \text{BESS_ADP} \text{ if BESS_SoC} = 100\% \mid \text{BESS_ACP} \text{ if BESS_SoC} = 0\% \mid \text{Min}(\text{BESS_ADP}, \text{BESS_ACP}) \text{ otherwise}$$

Testing and Corner Cases



Accuracy Testing

Perform controlled tests with forced imbalance or PCS power limits to validate Available Power metrics from BMS and PCS.



PCS Idle & Standby Modes

Review all PCS operating modes and define how Power Availability is handled in standby or idle state.



High & Low SoC Estimation Error

Special consideration may be taken in form of depth-weighted availability derate for derates at high and low BESS SoC.



SoC override to force SoC balancing

BESS SoC might be artificially lowered or raised to force BESS to continue charging or discharging.

What's Coming Next

We've covered the **Power Availability**.

In our next sessions, we'll dive deeper into **Energy Availability**.

Stay tuned for more Performance Engineering insights!

